

TABLE III
Touch Technologies and Their Characteristics

Technology	4-wire resistive	5-wire resistive	Capacitive
Durability	Three years	Five years	Two years
Stability	High	High	OK
Transparency	Bad	Bad	OK
Installation	Built-in/on-wall	Built-in/on-wall	Built-in
Touch	Anything	Anything	Conductive
Light-resistant	Good	Good	Bad
Response time	<10ms	<15ms	<15ms
Following speed	Good	Good	Good
Excursion	No	Big	Big
Monitor option	CRT or LCD	CRT or LCD	CRT or LCD or LED
Waterproof	Good	Good	Good

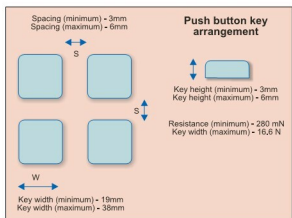


Fig. 5: Design input for a push button

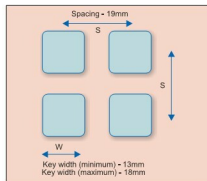


Fig. 6: Design input for a touchscreen

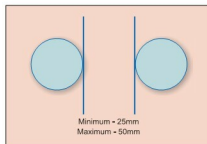


Fig. 7: Design input for spacing between knobs

classified into four major types:

Dials. These have graduated scales on which the indication of a value is shown by a pointer. An advantage of the dials is that these give the relative status of the product by showing where the needle is, as compared to full scale.

Warning devices. Warning displays call for attention and require action. For example, a red traffic light means that you must stop your vehicle.

Indicators. These displays have no graduated scales, but display text or numeric information, to show the state of a system. A good example is a digital panel meter, which shows the reading but never tells the maximum or minimum, unless explicitly mentioned.

Counters. Counters show the information directly as numbers.

Based on their functionality, displays can be grouped into three major categories:

Quantitative displays.

1. These display exact information. Digital quantitative displays present information directly as numbers. An example is a digital clock.

2. Analogue quantitative displays can also be used where the length or angle represents the information. For example, a thermometer where the column of mercury or alcohol represents temperature.

3. Use of a particular quantitative display depends on the kind of information that is required. If you need a precise reading, then digital indicators are most easily read.

Qualitative displays.

1. These displays give information about particular states, for example, hot or cold, alarm or no alarm, etc. These can provide information about the rate of change or direction of deviation from a desired value.

2. These displays may include indicators and warning devices, and can be used in circumstances where you only need to know whether a certain condition exists. For example, when temperature of

move the hand away from the keyboard; a stylus also requires hand movements (to use the pen)

3. **Fatigue:** Could strain arm muscles after heavy use, especially if the screen is placed vertically. Also, the user has to sit/stand close to the screen

4. **Dirt:** The screen gets dirty from fingerprints

5. **Screen coverage:** The user's hand, finger or pen may obstruct parts of the screen

6. **Activation:** Usually direct activation of the selected function, when the screen is touched; there is no special activation button as in the case of a light pen or mouse

Some design inputs, especially the spacing between different types of keys to be used, are given in Fig. 5 for a push button, Fig. 6 for a touchscreen and Fig. 7 for spacing between knobs, for designers to follow.

Let us now look at displays, which are an integral part of the UI. A UI is complete only when input (keys) and output (displays) go together.

Some displays used in products and design inputs for these are discussed next.

Displays and output devices. In an electronic system, displays can be